Daily Combinatorics Drill #5

Sheet by David Akinyele-Aje

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Stars & Bars · Substitution Shifts · Inclusion–Exclusion · Counting with Caps.

Section A Rapid Recognition ($\leq$15 s each)

1. Non-negative (x, y) with $x + y = 10$?

Ans: 11

2. 5 identical sweets to 3 children (zeros allowed)?

Ans: $\binom{7}{2} = 21$

3. $|A \cup B|$ from $15, 12, 5$?

Ans: $15 + 12 - 5 = 22$

4. Multiples of 3 up to 100?

Ans: $\lfloor 100/3 \rfloor = 33$

5. $|A| = 8$, $|B| = 6$, $|A \cup B| = 11$: find $|A \cap B|$.

Ans: $8 + 6 - 11 = 3$

6. Non-negative triples, $x + y + z = 4$?

Ans: $\binom{6}{2} = 15$

7. Positive triples, $x + y + z = 6$?

Ans: $\binom{5}{2} = 10$

8. Class of 30: 20 French, 15 German, 8 both — neither?

Ans: $30 - (20 + 15 - 8) = 3$

9. True or false: 7 has six ordered two-part sums?

Ans: True $(1{+}6,\ 2{+}5,\ \ldots,\ 6{+}1)$

10. 4 identical marbles, 2 boxes?

Ans: 5

Section B Manipulation Drills ($\leq$50 s each)

1. Non-negative triples, $x + y + z = 12$?

Ans: $\binom{14}{2} = 91$

2. Positive triples, $x + y + z = 12$?

Ans: $\binom{11}{2} = 55$

3. 10 sweets, 4 children, each at least one?

Ans: $\binom{9}{3} = 84$

4. 1–1000, divisible by neither 2 nor 5?

Ans: $1000 - 500 - 200 + 100 = 400$

5. Non-negative, $x + y + z = 10$, $x \geq 2$?

Ans: $\binom{10}{2} = 45$

6. Arrangements of $1, 2, 3, 4$ with 1 not first?

Ans: $4! - 3! = 18$

7. 1–100, divisible by 2 or 3?

Ans: $50 + 33 - 16 = 67$

8. Ordered three-part sums of 8?

Ans: $\binom{7}{2} = 21$

9. Non-negative (x, y) with $x + y \leq 5$?

Ans: $\binom{7}{2} = 21$

10. 5-letter A/B strings using both letters?

Ans: $2^5 - 2 = 30$

Section C Substitution & Structure ($\leq$90 s each)

1. Positive quadruples, $x + y + z + w = 15$, none exceeding 5?

Ans: 52

2. Non-negative triples, $x + y + z = 15$, each ≤ 6?

Ans: 10

3. Tea/coffee/juice: how many like none?

Ans: $100 - 80 = 20$

4. 1–1000, divisible by 2, 3, or 5?

Ans: 734

5. 8 apples, 3 children, each at most 4?

Ans: 15

6. ABCDE arrangements: A not first, B not second?

Ans: $120 - 24 - 24 + 6 = 78$

7. Three dice summing to 10?

Ans: 27

8. Positive integers below 1000 with digit sum 12?

Ans: $\binom{14}{2} - 3\binom{4}{2} = 91 - 18 = 73$

Section D Challenge (SMC / BMO1 difficulty)

1. Show exactly 9 of the 24 letter placements have no letter correct.

Ans: $D_4 = 9$

2. Prove: compositions of n number 2^{n-1}.

Ans: Proof below.

3. Integers 1–360 coprime to 360?

Ans: 96

4. Five students, three clubs, none empty: show 150.

Ans: $3^5 - 3 \cdot 2^5 + 3 \cdot 1^5 = 150$

5. Eight identical sweets are shared among 3 children. Consider statements: (A) the first child gets at least 2 sweets, (B) exactly two children receive at least one sweet, (C) no child receives more than 4 sweets. Exactly one of the three statements is true. How many distributions satisfy this? *(after TMUA Practice Paper 2 Q4)*

Ans: 16